Crime Patterns in the Capital: Exploring Crime Patterns Across Washington D.C. in 2025

Temiloluwa Owoade

College of Arts and Sciences

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Dr Moussa Doumbia

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Introduction

A city of startling contrasts, Washington, D.C. is home to prestigious universities like Howard University, a seat of government authority, and a town dealing with ongoing public safety issues stemming from its complicated social and economic past. In this situation, comprehending crime is not only an intellectual endeavor; it is essential to developing policies that fairly benefit all citizens.

The Crime Incidents in 2025 dataset, which was obtained from DC Open Data (opendata.dc.gov), was utilized in this investigation. Each row represents a single crime incidence that was reported in the District during the 2025 calendar year (DC Open Data, 2025). Geographic coordinates (latitude and longitude), the kind of infraction, the date and time of occurrence and reporting, and a number of administrative categories, such as ward, police district, Police Service Area (PSA), neighborhood cluster, ANC, census tract, and voting precinct, are all recorded in the dataset.

This dataset's simultaneous three-dimensional lens, spatial (where crimes occur), temporal (when they occur), and categorical (what kinds of offenses are reported), makes it very compelling. Because of its multidimensionality, it can be used for a variety of analytical methods, such as temporal trend analysis, geographic hotspot mapping, and straightforward frequency tabulations.

This dataset has significant social value in addition to its technical significance. Urban equity, institutional trust, and resource allocation are structural issues that are intertwined with crime patterns. Analyzing the differences in offense kinds and rates between wards and time periods provides a valuable view into how structural and environmental elements influence D.C.'s everyday safety environment. residents, therefore this research is both data science and sociological analysis.

Objective

The purpose of this research is to investigate the categorical, temporal, and spatial aspects of recorded crime occurrences in Washington, D.C. in 2025. Six main research issues are addressed by the analysis using the DC Open Data Crime Incidents dataset:

- In Washington, D.C., what categories of offenses are most commonly recorded in 2025?
- What is the distribution of crime among the eight wards in the city, and which wards are most affected?
- Is the makeup of crime types comparatively consistent throughout the city, or does it differ considerably by ward?
- Do particular police shifts (day, evening, midnight) increase the likelihood of various crimes?
- Do crime rates in 2025 exhibit a seasonal pattern?
- Does the time it takes for crimes to be recorded vary depending on the type of offense?

When combined, these inquiries provide an organized investigation into the underlying behavioral and geographic trends that influence public safety in the District as well as the surface-level statistics. The ultimate objective is to offer data-driven insights that can guide resources.

Materials and Methods

Material

- Jupyter Notebook (Anaconda)
- Packages used: Seaborn, Matplotlib, Numpy, Pandas, Plotly.express, Plotly.graph_objects
- Final Project Guidelines
- DC Crime Incident 2025 report: Crime_Incidents_in_2025.csv

The Crime Incidents in 2025 dataset contained 24 columns: case control number, report date, start date, end date, block location, offense type, method of offense, policing shift, ward, advisory neighborhood commission, police district, police service area, neighborhood cluster, block group, census tract, voting precinct, latitude, longitude, object ID, duration in days, month, month name, week, and day.

Method

Data Preparation & Cleaning Summary

Using pandas and geopandas, the raw dataset was refined through the following steps to ensure analytical integrity:

1. Data Cleaning & Missing Values
   - Removal: Dropped columns with excessive nulls (e.g., BID) and redundant, collinear coordinates (e.g., XBLOCK, YBLOCK).
   - Geospatial Recovery: Instead of deleting records with missing DISTRICT or PSA data, a spatial join was performed. Crime coordinates were mapped against official DC shapefiles to accurately recover missing labels (DC Open Data, 2025)(DC Open Data, 2025).
   - Placeholders: Missing END_DATE values were set to a placeholder (January 1, 2026) to signify ongoing investigations without skewing duration statistics.
2. Formatting & Standardization
   - Date Conversion: REPORT_DAT, START_DATE, and END_DATE were converted from strings to datetime objects.
   - Categorical Uniformity: String fields (e.g., OFFENSE, METHOD) were trimmed of whitespace and standardized to uppercase.
   - Deduplication: Verified the dataset was free of duplicates using the unique OBJECTID.
3. Feature Engineering
   - Duration Analysis: Created DURATION_DAYS (difference between end and start dates) and flagged outliers using the Interquartile Range (IQR) method.
   - Temporal Decomposition: Derived MONTH, WEEK, and DAY from the report date to enable time-series analysis.

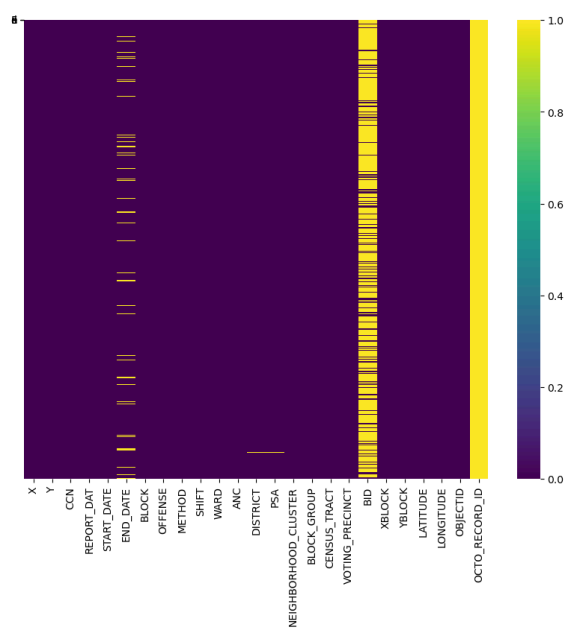

Figure 1: Dataset Before Cleaning The OCTO_RECORD_ID and BID columns shows large data gaps, justifying the deletion of the columns. Additionally, the smaller gaps observed under PSA, DISTRICT, and END_DATE resolved.

Storytelling

A clearer understanding of Crime in DC - Offense Distribution

Every offense type was sorted by frequency throughout the dataset using a horizontal count plot. With 11,643 incidents, more than twice as many as the next most frequent infraction,theft/other emerged as the most prevalent offense category. Theft from Auto (5,366) and Motor Vehicle Theft (3,933) came next, demonstrating that property and vehicle-related offenses account for the vast bulk of D.C.'s recorded crime volume. On the other end of the range, severe crimes like sexual abuse (103 cases) and homicide (127 incidents) happened much less frequently. This distribution makes a fundamental point right away: in terms of raw volume, property crime has affected D.C. significantly more than violent crime.

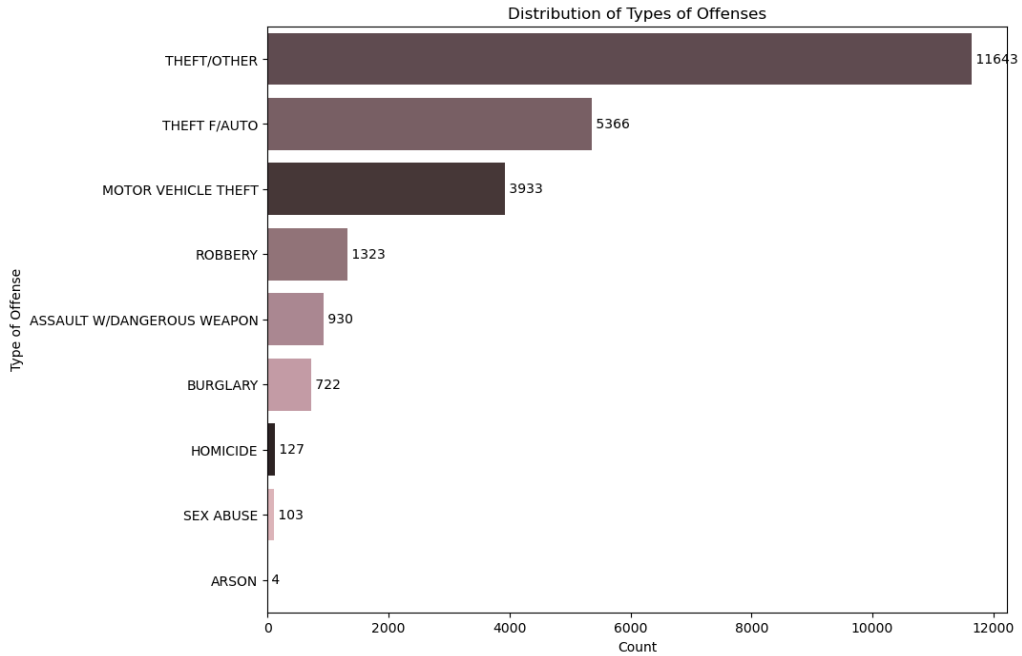

Figure 2: Distribution of Types of Offenses Theft/Other is the most frequent offense by a wide margin (11,643 incidents). Theft from Auto (5,366) and Motor Vehicle Theft (3,933) follow, indicating that vehicle-related property crimes are a significant portion of the city's total crime volume. Violent Crimes, such as Homicide (127) and Sex Abuse (103), occur at much lower frequencies compared to property-related offenses.

Which Neighborhoods Bear the Burden? - Ward-Level Crime Distribution

Crime is not uniformly spread throughout the District, according to a bar chart of incident numbers by ward. The biggest number of occurrences were reported in Wards 2 and 5, indicating that increased police presence may be necessary in these areas. In contrast, Ward 3 recorded the fewest incidents, which is in line with its image as a higher-income, primarily residential region. These trends show not only actual crime but also commercial activity, population density, and the level of interaction between locals and law enforcement. The results present a significant query on the fair distribution of resources among the eight administrative wards of the city.

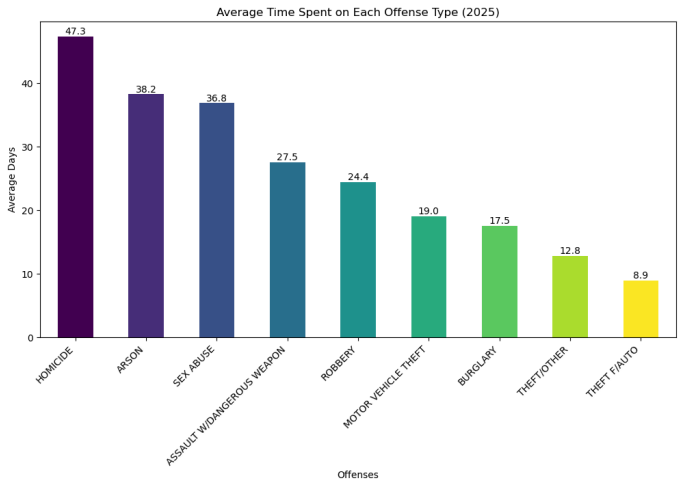

Figure 3: Average Time Spent on Each Offense Type Homicides and Sex Abuse represent the heavy lifting of the legal and investigative system, requiring roughly 5x more time per case than a standard vehicle theft.There is a clear inverse relationship between how often a crime happens and how long it takes to be resolved. This may be due to late reporting or complex cases.

Is Every Ward Affected by the Same Crimes? - Offense-Ward Heatmaps

Theft is most prevalent in Wards 1 and 2, hubs of high foot traffic, while violent crimes like homicide and assault are concentrated in Wards 7 and 8. Normalizing this data to account for crime volume confirms these distinct spatial patterns, with theft peaking in Ward 2 and violent offenses in Ward 8. Consequently, a "one-size-fits-all" policing strategy is ineffective. The District requires specialized approaches: high-visibility deterrence to combat theft in Ward 2, and increased investigative funding to address violent crime in Ward 8.

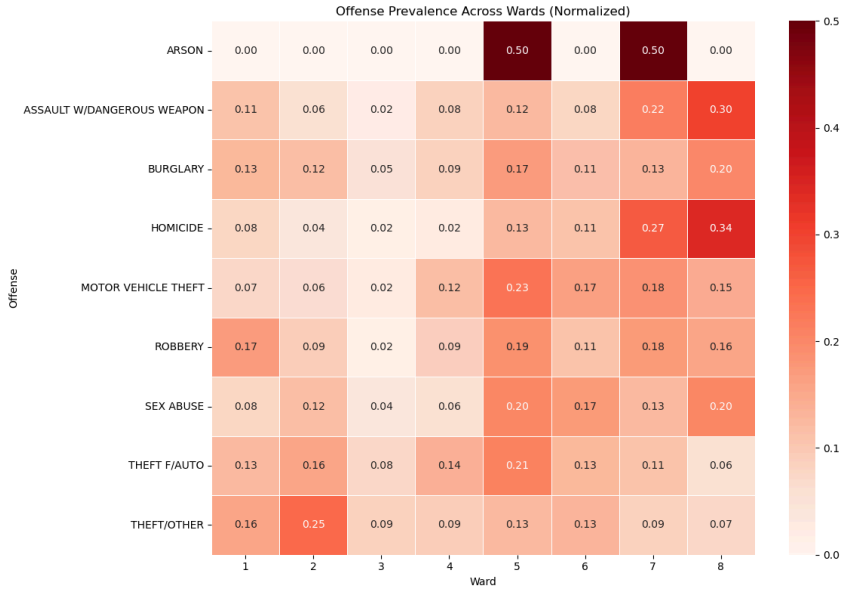

Figure 4: Offense Distribution Across DC Wards. This heatmap examines the joint distribution of offense type and ward across Washington D.C. in 2025. By encoding incident counts as color intensity, we can simultaneously compare crime composition across all eight wards and identify whether certain offenses are geographically concentrated or uniformly distributed, a pattern that individual ward-level bar charts would make difficult to compare at a glance.

When Does Crime Happen? – Temporal Shift Analysis

Reporting data shows that Day and Evening shifts record the highest incident volumes, while the Midnight shift records the fewest. However, these totals are likely skewed by reporting delays; many crimes occurring overnight are not reported until the following morning. A breakdown by offense type reveals a clear divide: property thefts are disproportionately reported during the day, whereas violent crimes (assault and robbery) are distributed more evenly across

all three shifts. This suggests that while the Midnight shift has lower volume, the severity of its incidents often outweighs the daytime totals.

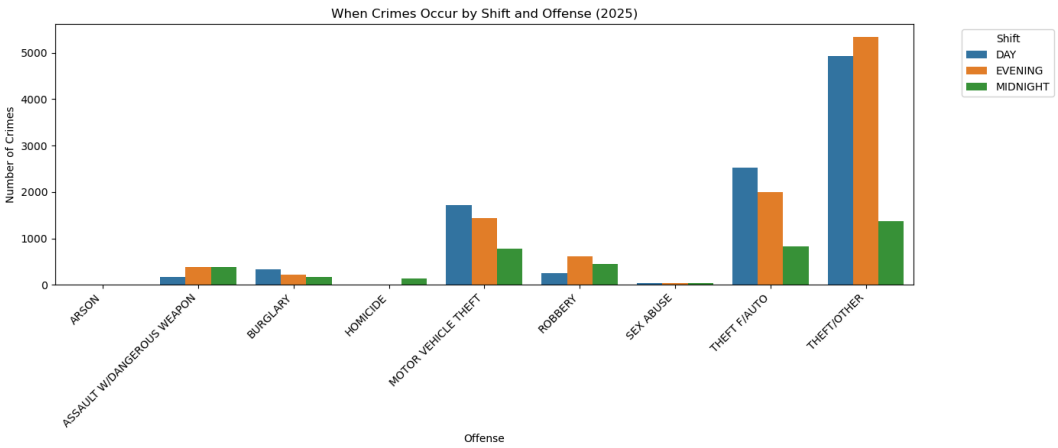

Figure 5: When Crimes Occur by Shift and Offense. The day shift is actually the most reported time for property theft, despite the fact that the "safe" sensation of daylight is statistically genuine for violent crimes like assault and homicide. However, this only shows the reported incidents, not necessarily the time they really occurred.

<u>Monthly and Seasonal Trends: Is There a Crime Season?</u>

A clear "summer surge" is confirmed by data from 2025: crime totals increased during the spring, peaked in July and August, and then significantly decreased as winter drew near. Longer daylight hours and more outdoor exercise are consistent with this pattern.This seasonality is driven by property crimes, according to the data. "Theft/Other" peaked in August with over 1,100 incidents, with vehicle-related thefts following a similar arc. On the other hand, violent crimes like assault and robbery were mostly unchanged, indicating that seasonal variations have less of an impact on them.

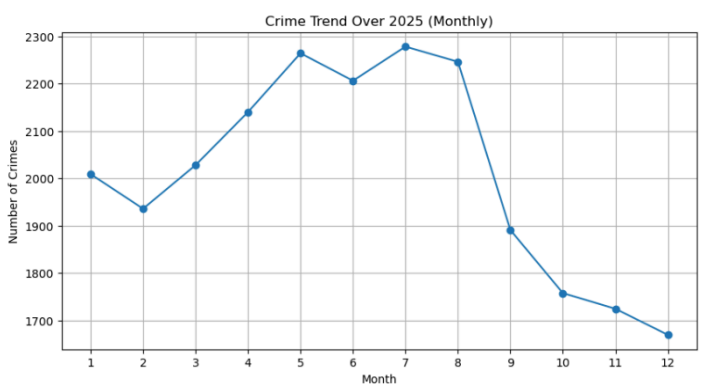

Figure 6: Monthly and Seasonal Trends. During the warmer months, the number of crimes increase and we see much less reporting during the colder months, especially December.

<u>Other Areas of Research:</u>

Analysis of high-severity, low-volume offenses reveals contrasting patterns. Homicide occurred consistently throughout 2025, peaking in February (17) and May (18) before declining in the fall. In contrast, Arson was rare and sporadic, appearing in only four months and never exceeding two incidents per month. Visualizing these crimes separately is essential to prevent their distinct trends from being buried by high-volume data.

Additionally, mapping incidents by latitude and longitude provides a clear spatial portrait of the city:

- Theft: The most ubiquitous crimes, heavily concentrated in the central business district and Northwest quadrant.

- Violent Crime: Assault and robbery clustered primarily in the northeastern and southeastern sectors.
- Transit Corridors: Motor vehicle thefts frequently followed major transit routes.
- High-Density Hubs: Neighborhood Clusters 25 and 8 are the city's highest-volume areas. Notably, the area surrounding Howard University sits at the intersection of several active clusters, marking it as a high-incident zone.

While the median reporting lag is 0.0 days (indicating most crimes are reported immediately), the mean lag is 14.6 days. This discrepancy is driven by a small subset of cases:

- Property Crimes: Usually reported and closed rapidly.
- Homicides: Average 47.3 days per case, reflecting their extreme complexity.

This paradox highlights a critical resource challenge: the most common crimes are the fastest to resolve, while the rarest crimes demand the most significant long-term investigative investment.

Conclusion

This analysis demonstrates that crime in Washington, D.C. is structured rather than random, following predictable spatial, temporal, and categorical patterns. A primary takeaway is the "Volume vs. Complexity" paradox: while high-frequency property crimes like theft drive total incident counts and are resolved quickly, rare but severe offenses like homicide require significantly more investigative time, averaging 47.3 days per case. Geographically, the city exhibits distinct "specializations," with Ward 2 serving as a hub for retail theft, Ward 5 for vehicle offenses, and Ward 8 for concentrated violent crime. Hyper-concentrated hotspots were also identified in Neighborhood Clusters 25 and 8, particularly along major transit and commercial corridors near Howard University.

Temporally, the District follows a clear "summer surge" rhythm, with incidents peaking in July and August due to increased outdoor activity. Most crimes are reported during the Day and Evening shifts, though the Midnight shift handles a disproportionate share of severe violent offenses despite its lower overall volume. While the 0.0-day median reporting lag confirms that most incidents are reported in real-time, the 14.6-day mean highlights discovery delays common in property crimes. Ultimately, these findings advocate for a tailored public safety strategy that balances high-visibility deterrence in commercial zones with deep investigative investment in residential corridors.

References

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